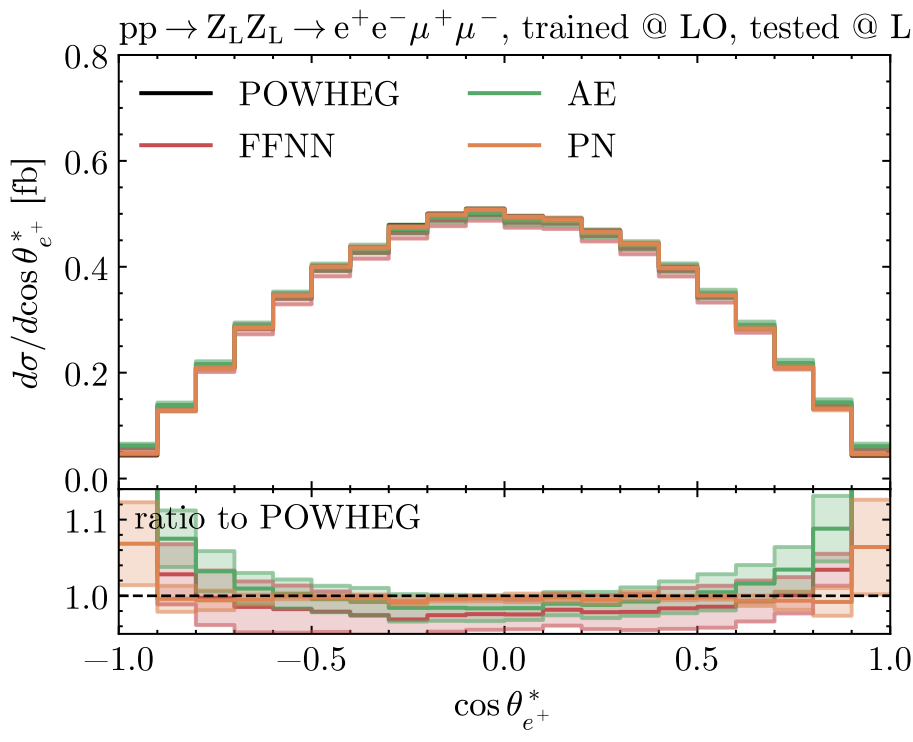
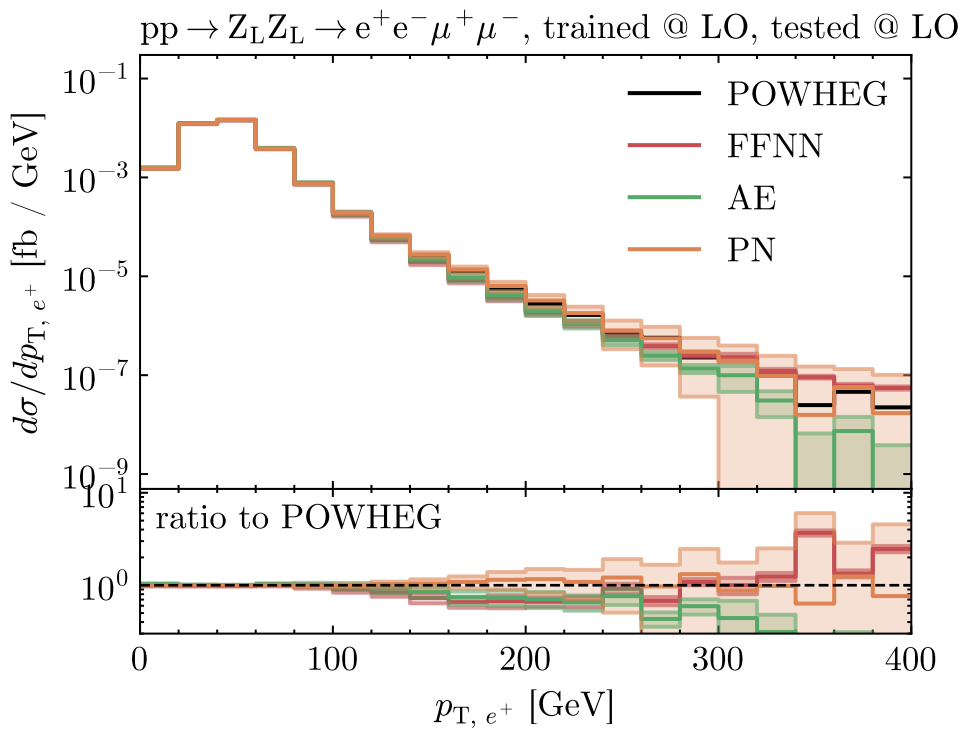
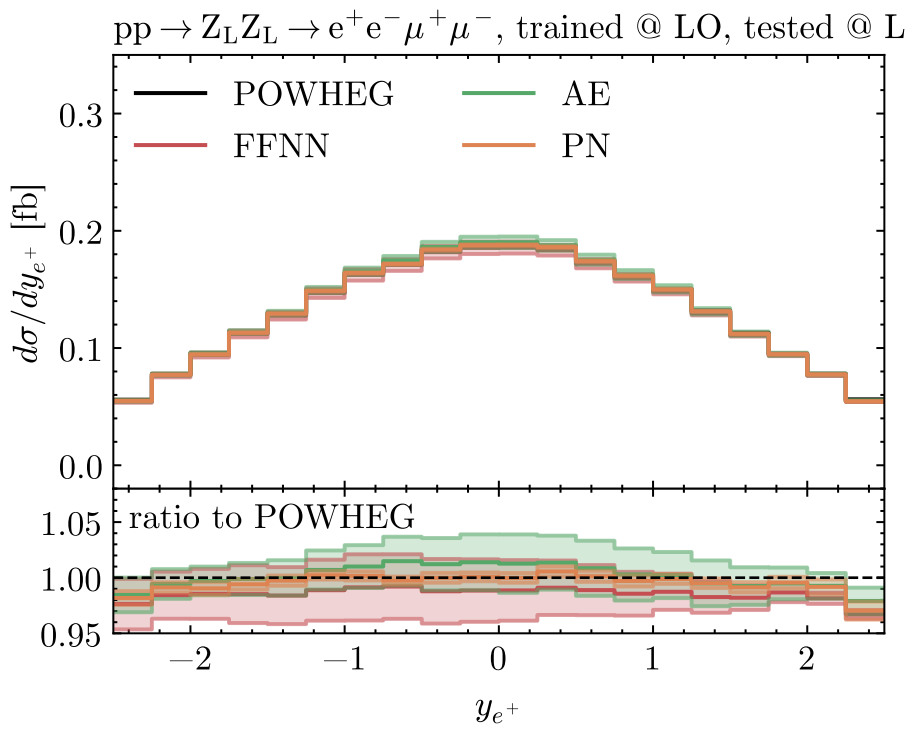
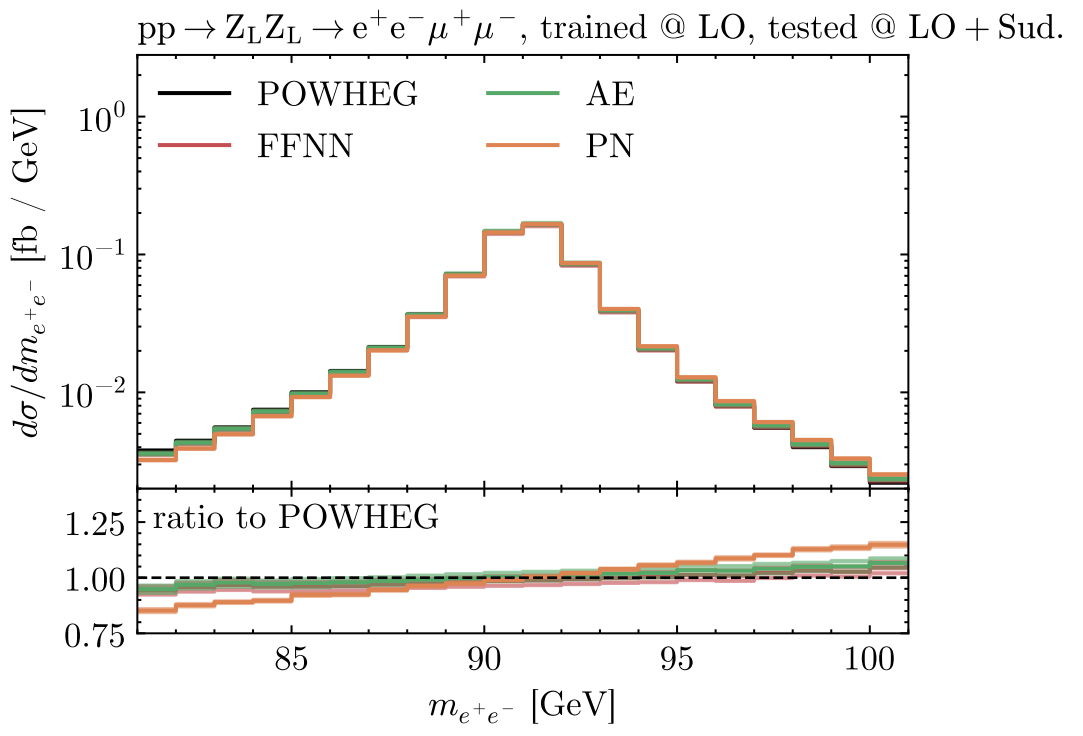
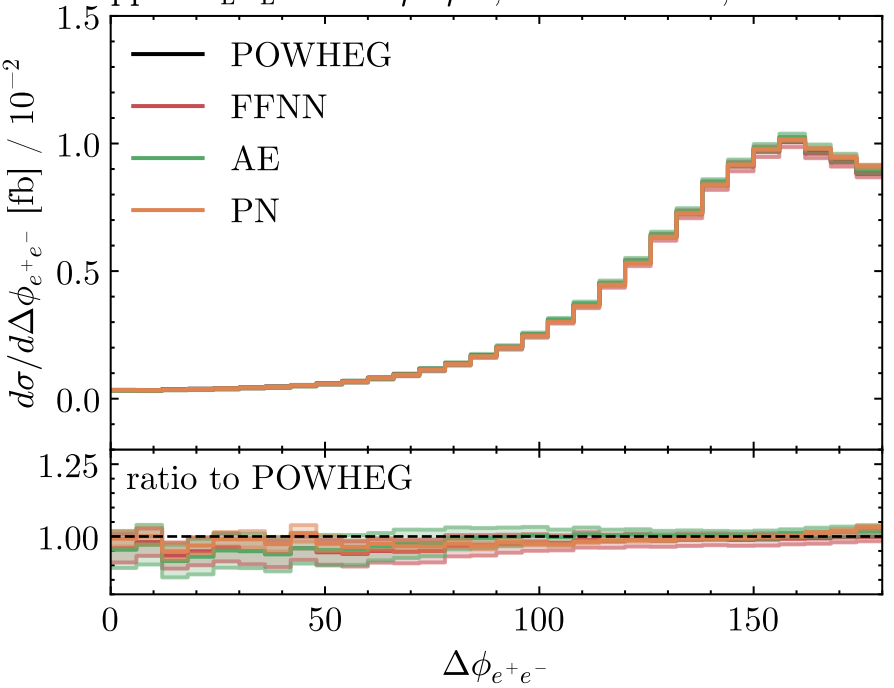
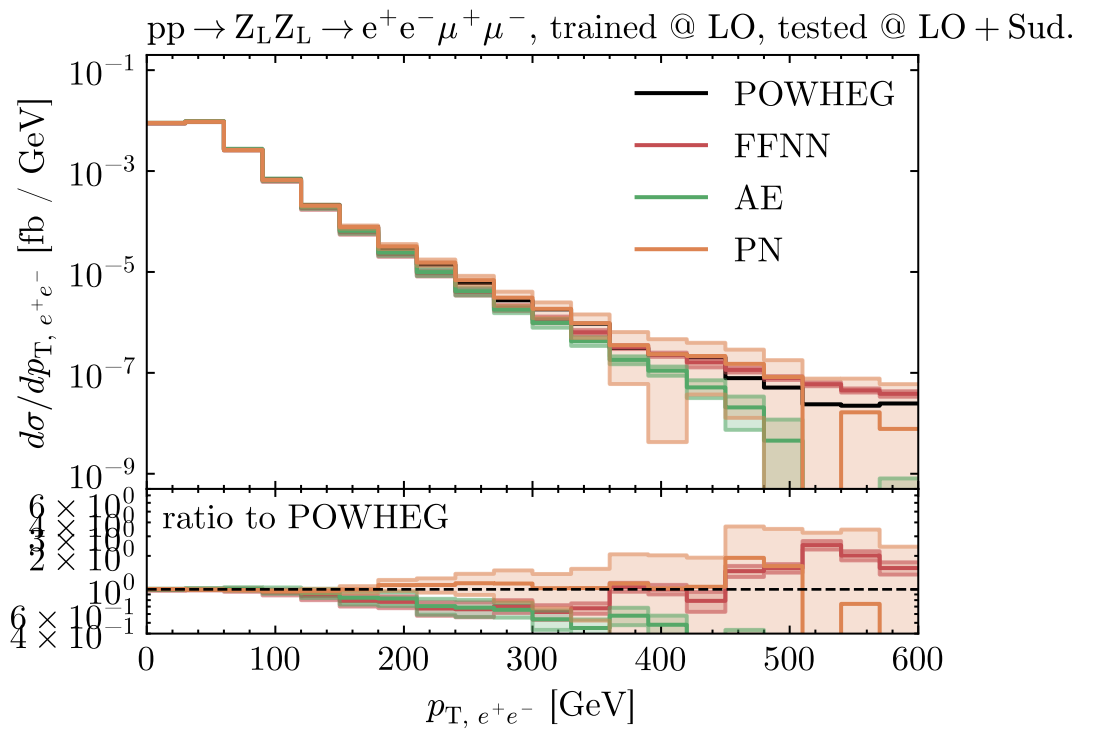

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.8
0.6
0.4
0.2
0.0
ratio to POWHEG
1.1
1.0
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

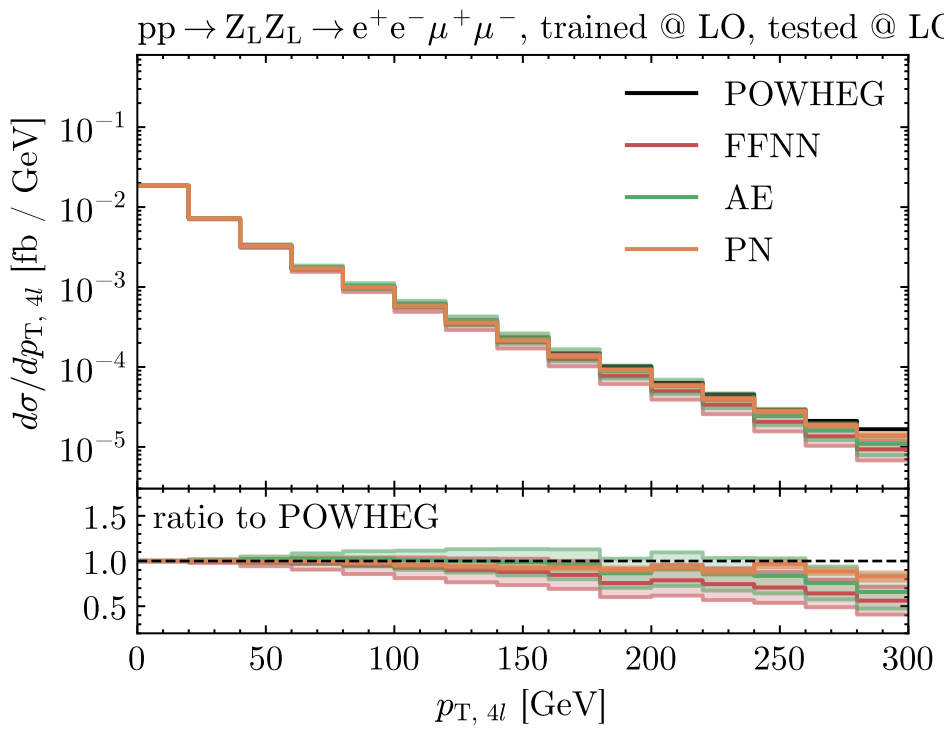

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dp_{T,\,e^+}$ [fb / GeV]
ratio to POWHEG
$p_{T,\,e^+}$ [GeV]

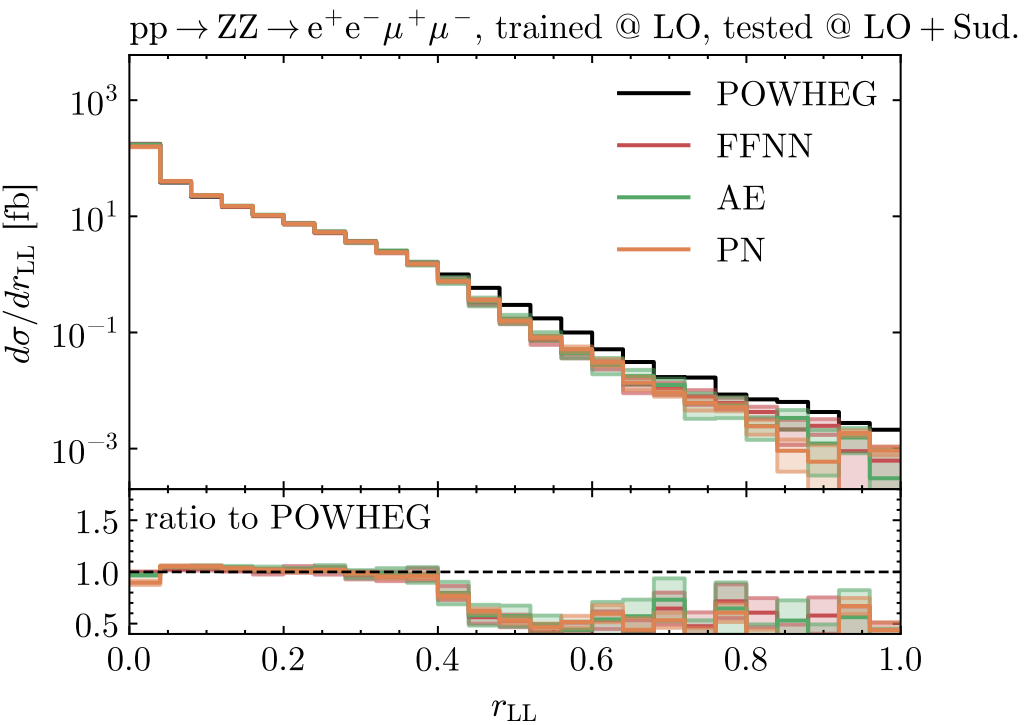

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dy_{e^+}$ [fb]
0.0
0.1
0.2
0.3
ratio to POWHEG
0.95
1.00
1.05
−2
−1
0
1
2
$y_{e^+}$


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
dσ/dm_{e⁺e⁻} [fb / GeV]
10^0
10^-1
10^-2
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
m_{e⁺e⁻} [GeV]


$pp \to Z_L Z_L \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/d\Delta\phi_{e^+e^-}$ [fb] / $10^{-2}$
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0
50
100
150
$\Delta\phi_{e^+e^-}$


$pp \to Z_L Z_L \to e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dp_{T,\,e^+e^-}$ [fb / GeV]
ratio to POWHEG
$p_{T,\,e^+e^-}$ [GeV]


pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dp_{T,\,4l}$ [fb / GeV]
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
$p_{T,\,4l}$ [GeV]


pp → ZZ → e⁺e⁻μ⁺μ⁻, trained @ LO, tested @ LO + Sud.
POWHEG
FFNN
AE
PN
$d\sigma/dr_{\mathrm{LL}}$ [fb]
$10^{3}$
$10^{1}$
$10^{-1}$
$10^{-3}$
ratio to POWHEG
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
$r_{\mathrm{LL}}$